A CORRESPONDENCE FOR THE GENERALIZED HECKE ALGEBRA OF THE METAPLECTIC COVER $\overline{SL(2,F)}$, F p-ADIC

DRAFT 1. CORRECTED AND EXPANDED VERSION

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ABSTRACT. This corrected and expanded version was prepared with the help of AI systems (Argon, ChatGPT and Claude). The Acknowledgments give details. We prove, using a technique developed for $GL(n)$ in Howe and Moy [HM85], a bijection between generalized Hecke algebras of $G = SL(2,F)$ over a p-adic field and those of its n-fold metaplectic cover $\overline{G}$. This result implies that there is a canonical correspondence between irreducible admissible representations of G and genuine irreducible admissible representations of $\overline{G}$ of "sufficiently large level" (depending on n, p).

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1. Introduction

This section introduces the covering group $\overline{G}$ studied in the paper and states what the paper proves. It ends with a running example and a description of how the paper is organized.

Let F be a p-adic field of characteristic 0, with uniformizer π, ring of integers O, and let q denote the cardinality of the residue field. Let $\mu_n(F)$ denote the group of nth roots of unity in F and assume $|\mu_n(F)| = n$. We will identify $\mu_n(F)$ with a subgroup of $\mathbb{C}^\times$ (via some fixed isomorphism $\theta : \mu_n(F) \to \mu_n(\mathbb{C})$). Write $\mu_n = \mu_n(F)$. Let $G = SL(2, F)$ and let $\mathfrak{g}$ denote its Lie algebra. Let $\overline{G} = \overline{SL(2, F)}$ denote the n-fold metaplectic cover defined by the Kubota cocycle

$$\beta(g_1, g_2) = \left(\frac{x(g_1 g_2)}{x(g_1)}, \frac{x(g_1 g_2)}{x(g_2)} \right),$$

for $g_1, g_2 \in G$, where

$$x\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} c, & \text{if } c \neq 0, \\ d, & \text{otherwise,} \end{cases}$$

and where $(\ , \ )$ is the nth power Hilbert symbol on $F^\times \times F^\times$. By §1 of Kubota [Ku67], β is a 2-cocycle (called a "factor set" in [Ku67]) which is trivial near the identity in $G \times G$, so $\overline{G}$ is a topological covering group of G. We denote elements of $\overline{G}$ by $\overline{x} = (x, \zeta)$, $x \in G$, $\zeta \in \mu_n$, and multiplication by

$$\overline{x}_1 \overline{x}_2 = (x_1, \zeta_1)(x_2, \zeta_2) = (x_1 x_2, \beta(x_1, x_2)\zeta_1 \zeta_2).$$

Let $\rho : \overline{G} \to G$ denote the natural surjection. If H is any subgroup of G then we define $\overline{H} = \rho^{-1}(H)$. We choose a Haar measure on G, to be normalized below, and a Haar measure on μ_n normalized so that $\mathrm{vol}(\mu_n) = 1$. Let the Haar measure on $\overline{G}$ be the product measure.

We prove, using a technique developed for $GL(n)$ in [HM85], a bijection between generalized Hecke algebras of G and those of $\overline{G}$. A precise statement will be given in Theorem 8.1. This result implies that there is a canonical one-to-one correspondence between irreducible admissible representations of G and genuine irreducible admissible representations of $\overline{G}$ of "sufficiently large level" (depending on n, p). For the precise statement, see Corollary 8.2.

Example 1.1 (The running example)**.** Throughout the paper we illustrate the definitions with one example. Take $F = \mathbb{Q}_3$, so that $\pi = 3$, $O = \mathbb{Z}_3$ and $q = 3$. Take $n = 2$, so that $\mu_2 = \{\pm 1\}$ and $\overline{G}$ is the classical two-fold metaplectic cover of $SL(2, \mathbb{Q}_3)$. The quadratic Hilbert symbol is $(a, b) = 1$ when a and b are both units, because $p = 3$ is odd. For a unit u it is $(u, 3) = \left(\frac{\bar{u}}{3}\right)$, the Legendre symbol of the residue $\bar{u} \in \mathbb{F}_3$. For example, $(-1, 3) = -1$, since -1 is not a square modulo 3. Here $\gcd(p, 2n) = \gcd(3, 4) = 1$, so every result of the paper applies, including those of Section 9. We use level $k = 1$ for the generalized Hecke algebras.

The paper is organized as follows. Section 2 reviews the literature. Section 3 explains the background on Hecke algebras, covering groups and representations. Section 4 fixes notation and defines the two Hecke algebras. Section 5 proves some lemmas on convolution of genuine functions. Sections 6 and 7 give generators and relations for the two algebras. Section 8 proves the main theorem and its consequences for representations. Section 9 discusses the Iwahori case. Section 10 summarizes the results, and Section 11 is a glossary.

2. Literature review

This section places the paper among related work. Section 2.1 covers work up to the original 1998 version of the paper. Section 2.2 covers later work on Hecke algebras of covering groups.

2.1. **Work up to 1998.** The paper builds on four tools. Kubota [Ku67] constructed the n-fold cover of $SL(2,F)$ by an explicit cocycle, which is the cocycle β used here. Kazhdan and Patterson [KP84] studied n-fold covers of $GL(r)$ and their genuine representations in general. Howe and Moy [HM85] determined generators and relations for the generalized Hecke algebras of $GL(n)$ with respect to filtration subgroups of the Iwahori subgroup. Our proof follows their method closely. Borel [Bo76] related representations with Iwahori-fixed vectors to modules over the Iwahori–Hecke algebra, using the induced and produced modules that we adapt in Section 8. Silberger's book [Si79] supplies the facts on matrix coefficients that we use there.

The study of genuine admissible representations on p-adic groups has its origins in the book [Ge76] by S. Gelbart. Flicker [Fl80] determined a comparison of Hecke algebras of *smooth functions* on $GL(2,F)$ and $\overline{GL(2,F)}$ using the matching of orbital integrals. This was then used, with the Arthur-Selberg trace formula, to establish a global "Shimura correspondence". Neither of these two works used the Hecke algebras considered here. G. Savin [Sa88] has given an isomorphism between the Iwahori-Hecke algebras in the case when p is relatively prime to $2n$ and G is any reductive p-adic group which splits over O. This isomorphism generalized a result of Flicker and Kazhdan [FK86] who considered the special case of $GL(r)$. (Savin's result is discussed briefly in §9 below.) The works of Rallis-Schiffmann [RS77] and Schultz [Sc98] also dealt with a correspondence between representations on $\overline{G}$ and on G (or a closely related group), though from a different perspective and using different methods. Our hope is that the result in the present paper will form one small step, along with Waldspurger's multiplicity one theorem for $\overline{SL(2,F)}$ [Wa80] and the Arthur-Selberg trace formula, in an eventual proof of Labesse's multiplicity one conjecture for $SL(2)$.

2.2. **Work after 1998.** Since 1998, Hecke algebras of covering groups have been studied with the theory of types and, more recently, with L-groups for covers. This subsection summarizes the work closest to the present paper.

In the same year as the original paper, Bushnell and Kutzko [BK98] developed the theory of types. A type is a pair consisting of a compact open subgroup and a representation of it, whose Hecke algebra controls one Bernstein component of the category of smooth representations. The algebra $\mathcal{H}(G//B_k)$ studied here is the Hecke algebra attached to the pair consisting of B_k and its trivial representation. This paper does not use the theory of types, and does not address whether that pair is a type.

Savin [Sa04] studied unramified genuine representations of more general covering groups. Gan and Savin [GS12] used types for the even and odd Weil representations of the metaplectic group $Mp(2n)$. They proved that the resulting Hecke algebras are isomorphic to Iwahori–Hecke algebras of odd orthogonal groups, when the residual characteristic is odd. Takeda and Wood [TW18] extended this to even residual characteristic. Chen and Li [CL25] revisited the Gan–Savin correspondence from the point of view of endoscopy. They showed that it preserves L-parameters. These results concern the Iwahori level, as in Section 9, rather than the deeper levels B_k of the present paper.

Karasiewicz [Ka21] gave a Bernstein presentation of a genuine Hecke algebra on the double cover of a simply laced Chevalley group over $\mathbb{Q}_2$. He used it to identify the algebra with an Iwahori–Hecke algebra of a linear group. Peskin [Pe13] studied genuine mod p representations of the metaplectic cover of $SL(2,F)$. She showed that its genuine mod p spherical Hecke algebra is isomorphic to that of $PGL(2,F)$.

Finally, Weissman [We18] defined an L-group for a broad class of covering groups, the Brylinski–Deligne covers. The volume containing his paper also has a historical introduction by Gan, Gao and

Weissman [GGW18]. In this framework the dual group of the n-fold cover of $SL(2)$ depends on the parity of n, in agreement with the description used in Section 9.

3. Background

This section explains the concepts the paper relies on, in more detail than the glossary. Each subsection ends with an example taken from the running example, Example 1.1. Readers who know the theory can skip to Section 4.

3.1. **Hecke algebras of totally disconnected groups.** This subsection explains why the generalized Hecke algebra of Definition 4.1 is the natural object for studying representations. Let H be a locally compact, totally disconnected, unimodular group, and let K be a compact open subgroup. A function on H that is locally constant, compactly supported and bi-K-invariant is a finite linear combination of characteristic functions of double cosets KhK. Convolution makes these functions into an associative algebra, and the characteristic function of K (suitably normalized) is its unit.

The reason for studying this algebra is the following. If (π, V) is a smooth representation of H, then the operator $\pi(f) = \int_H f(h)\pi(h)\,dh$ maps V into the space V^K of K-fixed vectors, for every f in the algebra. So V^K is a module over the algebra. For a large class of representations, the module V^K determines V. The precise statements are recalled in Proposition 8.9. So a comparison of Hecke algebras of two groups gives a comparison of their representations.

Example 3.1. Take $H = F^\times$ and $K = O^\times$. Every element of $F^\times$ is $\pi^m u$ with $m \in \mathbb{Z}$ and $u \in O^\times$, so the double cosets are the sets $\pi^m O^\times$. Let t_m be the characteristic function of $\pi^m O^\times$, and give $O^\times$ volume 1. Then $t_m * t_{m'} = t_{m+m'}$. So the Hecke algebra is the Laurent polynomial algebra $\mathbb{C}[t, t^{-1}]$, with $t = t_1$. A smooth character χ of $F^\times$ that is trivial on $O^\times$ acts on its one-dimensional space by $t \mapsto \chi(\pi)$. This identifies such characters with nonzero complex numbers.

3.2. **The Hilbert symbol and the Kubota cover.** This subsection explains the ingredients of the cover $\overline{G}$ defined in Section 1. The nth power Hilbert symbol $(\ ,\ )$ is a map $F^\times \times F^\times \to \mu_n$. It is multiplicative in each variable, satisfies $(a, b)(b, a) = 1$, and $(a, b) = 1$ when a or b is an nth power. Kubota [Ku67] used it to write down the explicit 2-cocycle β of Section 1. A 2-cocycle is exactly what is needed to make the set $G \times \mu_n$ into a group with the multiplication rule given there. The group $\overline{G}$ is a central extension of G by μ_n. It is not a matrix group in general.

Example 3.2. In the running example, take $a \in F^\times$ and $h(a) = \mathrm{diag}(a, a^{-1})$. For diagonal matrices the lower left entry is 0, so $x(h(a)) = a^{-1}$. For $a, b \in F^\times$ this gives

$$\beta(h(a), h(b)) = \left(b^{-1}, a^{-1}\right) = (b, a).$$

With $a = -1$ and $b = 3$ in $\mathbb{Q}_3$ we get $\beta(h(-1), h(3)) = (3, -1) = (-1, 3)^{-1} = -1$. So the cocycle takes the nontrivial value -1 on the diagonal torus. As a second example, $s_1^2 = -I$ and $x(s_1) = 1$, $x(-I) = -1$. So $\beta(s_1, s_1) = (-1, -1)$, which equals 1 in $\mathbb{Q}_3$ because -1 is a unit.

3.3. **The Iwahori subgroup and the Iwahori–Bruhat decomposition.** This subsection explains how the double cosets that index a basis of the Hecke algebra are organized. The Iwahori subgroup B of Section 4 consists of the matrices in $SL(2, O)$ whose lower left entry lies in πO. The affine Weyl group W_a is the infinite dihedral group generated by the images of s_1 and s_2. Each element w of W_a is a reduced alternating word in s_1 and s_2, and its length $\ell(w)$ is the number of letters. The Iwahori–Bruhat decomposition says that G is the disjoint union of the double cosets BwB, $w \in W_a$. The filtration subgroups B_k are normal in B, so each double coset $B_k g B_k$ is contained in some BwB. Remark 6.2 shows that $\mathrm{vol}(B_k b w B_k) = q^{\ell(w)}$ when $\mathrm{vol}(B_k) = 1$.

Example 3.3. The elements of W_a of length at most 2 are 1, s_1, s_2, s_1s_2 and s_2s_1. The element $t_0 = s_1^{-1}s_2 = h(\pi)$ of Section 4 has length 2. In the running example with $k = 1$, the formula above gives $\mathrm{vol}(B_1s_1B_1) = 3$ and $\mathrm{vol}(B_1s_1s_2B_1) = 9$. The index of B_1 in B is $q^2(q-1) = 18$. Indeed, B maps onto the upper triangular subgroup of $SL(2,\mathbb{F}_3)$, which has order $q(q-1) = 6$, and the kernel contains B_1 with index $q = 3$, coming from the lower left entry modulo π^2.

3.4. **Genuine representations, induced and produced modules.** This subsection explains the representation-theoretic notions of Section 8. A representation of $\overline{G}$ is **genuine** if μ_n acts through the fixed embedding $\mu_n \subset \mathbb{C}^\times$. Genuine representations do not factor through G, so they are the new representations that the cover contributes. A smooth representation is **admissible** if each space of vectors fixed by a compact open subgroup is finite dimensional. Its **matrix coefficients** are the functions $g \mapsto \langle v^*, \pi(g)v\rangle$. An irreducible admissible representation is **supercuspidal** if its matrix coefficients are compactly supported, and **square-integrable** if they are square-integrable. Definitions 8.4 and 8.10 make these notions precise for $\overline{G}$.

Borel [Bo76] constructed two ways back from Hecke modules to representations. Given a module W over $\mathcal{H}(H//K)$, the **induced** module is $C_c(H/K) \otimes_{\mathcal{H}(H//K)} W$. The **produced** module is the smooth part of $\mathrm{Hom}_{\mathcal{H}(H//K)}(C_c(K\backslash H), W)$. Proposition 8.9 recalls their properties in the genuine setting.

Example 3.4. Take $n = 1$, so that $\overline{G} = G$, and let $\mathbf{1}$ be the trivial representation of G. Its space of B_k-fixed vectors is all of $\mathbb{C}$. A function f in $\mathcal{H}(G//B_k)$ acts on it by the scalar $\int_G f(g)\,dg$. So f_g acts by $\mathrm{vol}(B_kgB_k)$. In the running example with $k = 1$, f_{s_1} acts by 3. The matrix coefficients of $\mathbf{1}$ are constant functions. They are not compactly supported, and not square-integrable, since G is not compact. So $\mathbf{1}$ is neither supercuspidal nor square-integrable.

4. Some notation

This section fixes the matrices, subgroups and Hecke algebras used in the rest of the paper. It begins with a table of notation. It then defines the generalized Hecke algebra of G (Definition 4.1) and its genuine analog for $\overline{G}$ (Definition 4.5).

Symbol	Meaning
$F,\ \pi,\ \mathcal{O},\ q$	p-adic field, uniformizer, ring of integers, order of the residue field $\mathbb{F}_q$
$\mu_n,\ \theta$	n^{th} roots of unity in F, fixed isomorphism onto $\mu_n(\mathbb{C})$
$(\ ,\ )$	n^{th} power Hilbert symbol
$\beta,\ x(g)$	Kubota cocycle and the entry used to define it (Section 1)
$G,\ \overline{G},\ \rho$	$SL(2,F)$, its n-fold cover, the projection $\overline{G} \to G$ (in Section 8, ρ also denotes the right regular representation)
$\overline{H}$	$\rho^{-1}(H)$ for a subgroup H of G
$s_1,\ s_2,\ t,\ t_0$	the matrices defined below
$u_\pm(x),\ h(x)$	unipotent and diagonal matrices defined below
$B,\ B_k$	Iwahori subgroup and its kth filtration subgroup
$\mathfrak{g},\ \mathfrak{b},\ \mathfrak{b}_k$	Lie algebras of G, B, B_k
$U_k^\pm,\ T_k$	unipotent and diagonal parts of B_k (Lemma 6.3 and Remark 7.1)
$W_a,\ W$	affine and finite Weyl groups (Section 6)
S	the set $\{s_1, s_2\} \cup B$ indexing the generators

$\mathcal{H}(H//K)$	generalized Hecke algebra (Definition 4.1)
$\mathcal{H}(\overline{G}//B_k)_{gen}$	metaplectic generalized Hecke algebra (Definition 4.5)
$f_g,\ \overline{f}_g$	characteristic function of $B_k g B_k$, and its genuine analog (Section 5)
$\beta',\ s,\ s_K$	equivalent cocycle, splitting function, Kubota splitting
B_k^*	the split copy $\{(b,1) \mid b \in B_k\}$ of B_k in $\overline{B}_k$
$B^{(n)}$	the subgroup of B of Remark 7.1
$\mathcal{H}^\wedge$	equivalence classes of irreducible modules of an algebra $\mathcal{H}$

Let

$$s_1 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad s_2 = \begin{pmatrix} 0 & -\pi^{-1} \\ \pi & 0 \end{pmatrix}, \quad t = \begin{pmatrix} 0 & -1 \\ \pi & 0 \end{pmatrix}, \quad t_0 = \begin{pmatrix} \pi & 0 \\ 0 & \pi^{-1} \end{pmatrix}.$$

Note $t_0 = s_1^{-1} s_2$, $t \notin G$, and $t s_1 t^{-1} = s_2$. Although $t \notin G$, conjugation by t defines an automorphism of G of order 2 (since $t^2 = -\pi I$ is central in $GL(2,F)$).

Let

$$u_+(x) = \begin{pmatrix} 1 & x \\ 0 & 1 \end{pmatrix}, \quad u_-(x) = \begin{pmatrix} 1 & 0 \\ x & 1 \end{pmatrix}, \quad h(x) = \begin{pmatrix} x & 0 \\ 0 & x^{-1} \end{pmatrix}.$$

Let

$$B = \left\{ \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \pi\mathcal{O} & \mathcal{O} \end{pmatrix} \right\} \cap G$$

denote the Iwahori subgroup,

$$\mathfrak{b} = \left\{ \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \pi\mathcal{O} & \mathcal{O} \end{pmatrix} \right\} \cap \mathfrak{g}$$

denote its Lie algebra, and for $k > 0$ let

$$B_k = \left(I + t^{2k} \begin{pmatrix} \mathcal{O} & \mathcal{O} \\ \pi\mathcal{O} & \mathcal{O} \end{pmatrix} \right) \cap G = \left\{ \begin{pmatrix} a & b \\ \pi c & d \end{pmatrix} \in G \;\middle|\; a-1,\ d-1,\ b,\ c \in \pi^k \mathcal{O} \right\}$$

denote the kth filtration subgroup of B. Note that $tBt^{-1} = B$ and $tB_k t^{-1} = B_k$. Let

$$\mathfrak{b}_k = \left\{ \begin{pmatrix} a & b \\ \pi c & -a \end{pmatrix} \;\middle|\; a, b, c \in \pi^k \mathcal{O} \right\}$$

denote the Lie algebra of B_k.

Definition 4.1. Let H denote a locally compact unimodular group and let K denote an open compact subgroup of H with Haar measure on H normalized so that K has measure one. Let $\mathcal{H}(H//K)$ denote the space of functions on H which are locally constant, compactly supported, and bi-K-invariant. This forms an algebra under convolution,

$$(f * g)(x) = \int_H f(y) g(y^{-1}x)\, dy,$$

called the **generalized Hecke algebra**.

Example 4.2. Example 3.1 is the case $H = F^\times$, $K = \mathcal{O}^\times$ of Definition 4.1. For the case used in this paper, take $H = G = SL(2,\mathbb{Q}_3)$ and $K = B_1$, as in the running example. The characteristic functions f_1, f_{s_1} and f_{s_2} of B_1, $B_1 s_1 B_1$ and $B_1 s_2 B_1$ lie in $\mathcal{H}(G//B_1)$. The function f_1 is the unit, and f_{s_1} and f_{s_2} are two of the generators of Proposition 6.1.

We shall study the algebra $\mathcal{H}(G//B_k)$ in Section 6.

Recall that a 2-cocycle $\alpha : H \times H \to \mu_n$ of a group H is **trivial** if there is a function $s : H \to \mu_n$ (called a **splitting**) such that $\alpha(h_1, h_2) = s(h_1)s(h_2)s(h_1h_2)^{-1}$, for $h_1, h_2 \in H$. In this case the cover of H defined by α is said to **split**, and the map $h \mapsto (h, s(h)^{-1})$ embeds H as a subgroup of $\overline{H}$. When α is replaced by the equivalent cocycle $\alpha'(h_1, h_2) = \alpha(h_1, h_2)s(h_1)^{-1}s(h_2)^{-1}s(h_1h_2)$ on $H \times H$, the splitting becomes trivial on H, and H embeds canonically as $H^* = \{(h, 1) \mid h \in H\} \subset \overline{H}$.

Example 4.3. For any function $s : H \to \mu_n$, the cocycle $\alpha(h_1, h_2) = s(h_1)s(h_2)s(h_1h_2)^{-1}$ is trivial, with splitting s. The cocycle α' obtained from it as above is identically 1. In the running example, $1 + 3\mathbb{Z}_3$ consists of squares in $\mathbb{Q}_3^\times$, by Hensel's Lemma. So the next paragraph applies with $k = 1$, and shows that $\overline{B}_1$ splits.

By Hensel's Lemma, if $k > 2\,\mathrm{ord}_\pi(n)$ (and in particular for any $k \geq 1$ when $\gcd(p, n) = 1$), we have $1 + \pi^k O \subset (F^\times)^n$. For such k, every $u \in 1 + \pi^k O$ is an nth power in $F^\times$, so the nth power Hilbert symbol satisfies $(u, y) = 1$ for all $u \in 1 + \pi^k O$ and all $y \in F^\times$. By [Ku67], Theorem 2, the Kubota splitting function $s_K : G \to \mu_n$ defined by

$$s_K\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{cases} (c, d), & \text{if } cd \neq 0 \text{ and } \mathrm{ord}_\pi(c) \text{ is odd,} \\ 1, & \text{otherwise,} \end{cases}$$

trivializes β on the principal congruence subgroup of level π^k, and satisfies $s_K|_{B_k} \equiv 1$ since $d \in 1 + \pi^k O \subset (F^\times)^n$ for all elements of B_k. In fact, the definition of β and the Iwahori factorization $B_k = U_k^- T_k U_k^+$ imply directly that $\overline{B}_k$ splits.

Recall that a complex-valued function f on $\overline{G}$ is called **genuine** if $f(g, \zeta) = \theta(\zeta) f(g, 1)$, for $g \in G$ and $\zeta \in \mu_n$. Here θ is a fixed isomorphism $\theta : \mu_n \to \mu_n(\mathbb{C})$. It being agreed that we may identify μ_n and $\mu_n(\mathbb{C})$, we shall, *from this point on, drop θ from the notation.*

Example 4.4. In the running example $n = 2$ and $\mu_2 = \{\pm 1\}$. A function f on $\overline{G}$ is genuine when $f(g, -1) = -f(g, 1)$ for all $g \in G$. So a genuine function is determined by its values on the elements $(g, 1)$, and changes sign on the other sheet of the cover. A nonzero function that factors through G, such as a nonzero constant, is never genuine when $n \geq 2$.

Definition 4.5. Assume that $k > 0$ is chosen so large that $1 + \pi^k O \subset (F^\times)^n$ (so in particular $\overline{B}_k$ splits). Fix a cocycle β' equivalent to β such that $\beta'|_{B_k \times B_k} = 1$, and identify B_k with the split subgroup $B_k^* = \{(b, 1) \mid b \in B_k\} \subset \overline{B}_k$. Assume that the Haar measure on $\overline{G}$ is normalized so that $\overline{B}_k$ has measure 1. Let $\mathcal{H}(\overline{G}//B_k)_{gen}$ denote the space of functions on $\overline{G}$ which are genuine, locally constant, compactly supported, and bi-B_k^*-invariant. This is the **(metaplectic) generalized Hecke algebra** on $\overline{G}$.

Example 4.6. In the running example, $k = 1$ satisfies the hypothesis of Definition 4.5, because $1 + 3\mathbb{Z}_3 \subset (\mathbb{Q}_3^\times)^2$. The function $\overline{f}_1$ defined by $\overline{f}_1(b, \zeta) = \zeta$ for $b \in B_1$, $\zeta = \pm 1$, and $\overline{f}_1 = 0$ off $\overline{B}_1$, is genuine and bi-B_1^*-invariant. It lies in $\mathcal{H}(\overline{G}//B_1)_{gen}$, and Proposition 7.2 shows that it is the unit of this algebra.

We will study this algebra in Section 7. Note that it depends on the splitting of the cocycle on B_k. To have a measure on $\overline{G}$ which fits with the measure on G used in §6, we shall also assume that μ_n has measure 1.

5. Some lemmas

This section collects the facts about double cosets and convolution that the proofs in Sections 6 and 7 need. The main point is Lemma 5.5. It shows that the basic product rule for characteristic functions of double cosets (Lemma 5.3) still holds for their genuine analogs.

Let $\mathbb{F}_q = O/\pi O$ denote the residue field of F.

Lemma 5.1. *For $k > 0$, $B_k/B_{k+1} \cong \mathfrak{b}_k/\mathfrak{b}_{k+1} \cong \mathbb{F}_q^3$, as abelian groups. (The second isomorphism is not canonical.)*

Proof. The graph of the first isomorphism is $\{(g \cdot B_{k+1}, X + \mathfrak{b}_{k+1}) \mid (I + X + \mathfrak{b}_{k+1}) \cap g \cdot B_{k+1} \neq \emptyset\}$ (namely, $I + X \bmod B_{k+1} \mapsto X \bmod \mathfrak{b}_{k+1}$ for $X \in \mathfrak{b}_k$). It is also a special case of a general result of Morris (Proposition 3.4 in [Mo88]). The second isomorphism is due to the fact that the surjective homomorphism

$$\begin{pmatrix} a & b \\ \pi c & -a \end{pmatrix} \longmapsto \left(\pi^{-k} a \bmod \pi,\ \pi^{-k} b \bmod \pi,\ \pi^{-k} c \bmod \pi\right) \in \mathbb{F}_q^3$$

has image $\mathbb{F}_q^3$ and kernel $\mathfrak{b}_{k+1}$. □

Example 5.2. In the running example with $k = 1$, Lemma 5.1 says that B_1/B_2 is an elementary abelian group of order $3^3 = 27$. The element $u_+(3) = I + X$ of B_1, with $X = \left(\begin{smallmatrix} 0 & 3 \\ 0 & 0 \end{smallmatrix}\right) \in \mathfrak{b}_1$, maps to $(0, 1, 0) \in \mathbb{F}_3^3$ under the second isomorphism, since $3^{-1} \cdot 3 = 1$.

The following result is stated as Proposition 2.2 in Chapter 3 of [HM85].

Lemma 5.3. *In the notation of Definition 4.1, let f_h denote the characteristic function of the double coset KhK in H. Then f_1 is the unit of $\mathcal{H}(H//K)$. More generally, if*

$$\mathrm{vol}(Kh_1K)\,\mathrm{vol}(Kh_2K) = \mathrm{vol}(Kh_1h_2K),$$

then $f_{h_1} * f_{h_2} = f_{h_1h_2}$.

Example 5.4. In Example 3.1, every double coset $\pi^m O^\times$ has volume 1. So the volume condition of Lemma 5.3 always holds, and the lemma gives $t_m * t_{m'} = t_{m+m'}$. In the running example, Remark 6.2 gives $\mathrm{vol}(B_1 s_1 B_1)\,\mathrm{vol}(B_1 s_2 B_1) = 3 \cdot 3 = \mathrm{vol}(B_1 s_1 s_2 B_1)$. So the lemma gives $f_{s_1} * f_{s_2} = f_{s_1 s_2}$. By contrast, the volume condition fails for $h_1 = h_2 = s_1$, since $s_1^2 = -I \in B$ and $\mathrm{vol}(B_1(-I)B_1) = 1 \neq 9$. In fact $f_{s_1} * f_{s_1} \neq f_{-I}$. By Lemma 6.3(d) the left side has integral 9, while the right side has integral 1. Relation A(i) of Proposition 6.1 gives the correct value.

We need a version of the previous lemma for genuine functions. Assume β' is chosen as in Definition 4.5 so that $\beta'|_{B_k \times B_k} = 1$. Since $1 + \pi^k O \subset (F^\times)^n$, for any $g \in G$ and any $b_1, b_2 \in B_k$ satisfying $b_1 g b_2 = g$, the compatibility of the Kubota splitting with $1 + \pi^k O \subset (F^\times)^n$ ensures that $(b_1, 1)(g, 1)(b_2, 1) = (g, 1)$. Thus, for each $g \in G$, there is a unique genuine bi-B_k^*-invariant function $\overline{f}_g$ supported on $\overline{B}_k(g, 1)\overline{B}_k$ satisfying $\overline{f}_g(g, 1) = 1$; explicitly, for $b_1, b_2 \in B_k$ and $\zeta \in \mu_n$,

$$\overline{f}_g\big((b_1, 1)(g, 1)(b_2, 1)(1, \zeta)\big) = \zeta,$$

and $\overline{f}_g(\overline{x}) = 0$ for $\overline{x} \notin \overline{B}_k(g, 1)\overline{B}_k$. When β' is additionally normalized on the double coset $B_k g B_k$ so that $\beta'(b_1, g)\beta'(b_1 g, b_2) = 1$ for $b_1, b_2 \in B_k$, this takes the simple form

$$\overline{f}_g(x, \zeta) = \begin{cases} \zeta, & x \in B_k g B_k, \\ 0, & \text{otherwise.} \end{cases}$$

Let

$$(\phi_1 * \phi_2)(g) = \int_{\overline{G}} \phi_1(x)\phi_2(x^{-1}g)\,dx, \quad g \in \overline{G}.$$

Now we prove a genuine analog of Lemma 5.3 above.

Lemma 5.5. *Assume* $\mathrm{vol}(\overline{B}_k) = 1$. *If* $g_1, g_2 \in G$ *are such that* $\beta'((g_1,1),(g_2,1)) = 1$ *on the double coset representatives and*

$$\mathrm{vol}(B_k g_1 B_k)\,\mathrm{vol}(B_k g_2 B_k) = \mathrm{vol}(B_k g_1 g_2 B_k),$$

then $\overline{f}_{g_1} * \overline{f}_{g_2} = \overline{f}_{g_1 g_2}$.

Proof. We compute

$$\begin{aligned}(\overline{f}_{g_1} * \overline{f}_{g_2})(g,1) &= \int_{\overline{G}} \overline{f}_{g_1}(x)\overline{f}_{g_2}(x^{-1}(g,1))\,dx \\ &= \int_G \overline{f}_{g_1}(x,1)\overline{f}_{g_2}(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx.\end{aligned}$$

Since $|\overline{f}_{g_i}(x,1)| = f_{g_i}(x)$, the support of this integral is contained in the support of $f_{g_1} * f_{g_2}$. By Lemma 5.3, $f_{g_1} * f_{g_2} = f_{g_1 g_2}$, so

$$\mathrm{supp}(\overline{f}_{g_1} * \overline{f}_{g_2}) \subset \mathrm{supp}(\overline{f}_{g_1 g_2}).$$

Since $\overline{f}_{g_1} * \overline{f}_{g_2}$ is genuine and bi-B_k^*-invariant, and the space of genuine bi-B_k^*-invariant functions supported on a single double coset $\overline{B}_k(g_1 g_2, 1)\overline{B}_k$ is one-dimensional (spanned by $\overline{f}_{g_1 g_2}$), we must have $\overline{f}_{g_1} * \overline{f}_{g_2} = c\overline{f}_{g_1 g_2}$ for some constant c. Evaluating at $(g_1 g_2, 1)$: by the volume condition $\mathrm{vol}(B_k g_1 B_k)\,\mathrm{vol}(B_k g_2 B_k) = \mathrm{vol}(B_k g_1 g_2 B_k)$, the map $B_k \times B_k \times B_k \to B_k g_1 g_2 B_k$ given by $(b_1, b_2, b_3) \mapsto b_1 g_1 b_2 g_2 b_3$ has fiber B_k at $g_1 g_2$, so for $x = g_1 b_2$ ($b_2 \in B_k$) we have $x^{-1} g_1 g_2 = b_2^{-1} g_2$ and the integrand at $g = g_1 g_2$ is identically 1 on $g_1 B_k$, yielding $c = \mathrm{vol}(g_1 B_k) = 1$. □

As in the above proof, we have the following expression for the convolution in terms of the cocycle.

Lemma 5.6. *For any genuine functions* $\phi_1, \phi_2 \in \mathcal{H}(\overline{G}//B_k)_{gen}$, *we have*

$$(\phi_1 * \phi_2)(g,\zeta) = \zeta \int_G \phi_1(x,1)\phi_2(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx,$$

for any $g \in G$ *and* $\zeta \in \mu_n$.

Note that the generalized metaplectic Hecke algebra depends on the cocycle β, so let us denote this dependence temporarily by $\mathcal{H}_\beta(\overline{G}//B_k)_{gen}$. If the cocycle is changed to an equivalent one, say to β', where

$$\beta(g,h) = \beta'(g,h)s(g)s(h)s(gh)^{-1}$$

is the cocycle modified by s and $s : G \to \mu_n$ is any function satisfying $s(1) = 1$ and $s|_{B_k} \equiv 1$ (for example, the Kubota splitting [Ku67, KP84]), then the two algebras are isomorphic.

Lemma 5.7. *Let* β, β' *be as above and let* $s : G \to \mu_n$ *satisfy* $s(1) = 1$. *There is a canonical isomorphism*

$$\phi : \mathcal{H}_{\beta'}(\overline{G}//B_k)_{gen} \xrightarrow{\sim} \mathcal{H}_\beta(\overline{G}//B_k)_{gen},$$

as algebras, defined by sending $f(g,\zeta) = \zeta f(g,1)$ *to* $\phi(f)(g,\zeta) = \zeta s(g) f(g,1)$. *In other words, under this mapping, we have*

$$\phi(f_1 *_{\beta'} f_2)(g,\zeta) = (\phi(f_1) *_\beta \phi(f_2))(g,\zeta),$$

where $*_\beta$ *denotes the convolution with respect to the* β *cocycle*

$$(\phi_1 *_\beta \phi_2)(g,\zeta) = \zeta \int_G \phi_1(x,1)\phi_2(x^{-1}g,1)\beta(x,x^{-1}g)^{-1}\,dx$$

and, similarly, $*_{\beta'}$ *denotes the convolution with respect to the* β' *cocycle.*

Proof. Using $\beta(x,x^{-1}g)^{-1} = \beta'(x,x^{-1}g)^{-1}s(x)^{-1}s(x^{-1}g)^{-1}s(g)$, we have

$$\begin{aligned}(\phi(f_1) *_\beta \phi(f_2))(g,\zeta) &= \zeta \int_G s(x) f_1(x,1) s(x^{-1}g) f_2(x^{-1}g,1)\beta(x,x^{-1}g)^{-1}\,dx \\ &= \zeta s(g) \int_G f_1(x,1) f_2(x^{-1}g,1)\beta'(x,x^{-1}g)^{-1}\,dx \\ &= \phi(f_1 *_{\beta'} f_2)(g,\zeta),\end{aligned}$$

as claimed. □

6. The structure of $\mathcal{H}(G//B_k)$

We want to determine the structure of the generalized Hecke algebra as a finitely generated algebra with generators and relations. Then we will do the same for the metaplectic analog and compare the two.

The affine Weyl group $W_a = N_G(T)/(T \cap B)$ of G is an infinite dihedral group generated by the images of s_1 and s_2, and the finite Weyl group W is generated by the image of s_1. Note that in G, $s_1^2 = s_2^2 = -I \in B$, and every element of the group $\langle s_1, s_2\rangle \subset G$ can be written up to sign $\pm I \in B$ as a reduced alternating word in s_1 and s_2 (in particular, $t_0 = s_1^{-1}s_2 = h(-1)s_1s_2$). We will use the Iwahori-Bruhat decomposition

$$G = BW_aB$$

to determine the structure of the generalized Hecke algebra.

Recall $(\mathrm{Ad}\, g)(x) = gxg^{-1}$. Let $S = \{s_1, s_2\} \cup B$.

Proposition 6.1. *The algebra* $\mathcal{H}(G//B_k)$ *is generated by the functions* f_g, $g \in S$. *Assume that* $\mathrm{vol}(B_k) = 1$. *These elements are subject to the relations below.*

(A) Relations for the s_i***'s (***$i = 1,2$***):***

(i) $f_{s_1} * f_{s_1} = c_1 \sum_x f_x$, *where* x *runs over a set of representatives of* $((-1)\cdot(\mathrm{Ad}\, s_1)(B_k)\cdot B_k)/B_k$ *and* $c_1 = \mathrm{vol}(B_k s_1 B_k) = q$,

(ii) $f_{s_2} * f_{s_2} = c_2 \sum_x f_x$, *where* x *runs over a set of representatives of* $((-1)\cdot(\mathrm{Ad}\, s_2)(B_k)\cdot B_k)/B_k$ *and* $c_2 = \mathrm{vol}(B_k s_2 B_k) = q$.

(B) Relations for elements of B***:***

(i) f_1 *is the identity of* $\mathcal{H}(G//B_k)$,

(ii) $f_x * f_y = f_{xy}$, *for* $x, y \in B$,

(iii) $f_{xg} = f_{gx} = f_g$, *for all* $x \in B_k,\ g \in G$.

(C) Mixed relations:

(i) $f_{s_i} * f_x = f_{\mathrm{Ad}\, s_i(x)} * f_{s_i}$, *for* $x \in B \cap (\mathrm{Ad}\, s_i^{-1})(B)$ *(*$i = 1,2$*),*

(ii) $f_{s_1} * f_{u_+(x)} * f_{s_1} = c_1^2 f_{\mathrm{Ad}\, s_1(u_+(x))} = c_1^2 f_{u_-(-x)}$, *for* $x \in \mathcal{O}^\times$, *where* $c_1 = \mathrm{vol}(B_k s_1 B_k)$,

(iii) $f_{s_2} * f_{u_-(-\pi x)} * f_{s_2} = c_2^2 f_{\mathrm{Ad}\, s_2(u_-(-\pi x))} = c_2^2 f_{u_+(x)}$, *for* $x \in O^\times$, *where* $c_2 = \mathrm{vol}(B_k s_2 B_k)$.

(The relations C(ii) and C(iii) must be expressed in terms of the generators S, as in [HM85]. *Using the Bruhat identity* $u_-(-x) = u_+(-x^{-1}) s_1 h(-x) u_+(-x^{-1})$ *for* $x \in O^\times$, *the relation corresponding to C(ii) is*

$$f_{s_1} * f_{u_+(x)} * f_{s_1} = c_1^2 \big(f_{u_+(-x^{-1})} * f_{s_1} * f_{h(-x)} * f_{u_+(-x^{-1})} \big).$$

Conjugating by t, *where* $t s_1 t^{-1} = s_2$, $t u_+(y) t^{-1} = u_-(-\pi y)$, *and* $t h(z) t^{-1} = h(z^{-1})$, *the relation corresponding to C(iii) is*

$$f_{s_2} * f_{u_-(-\pi x)} * f_{s_2} = c_2^2 \big(f_{u_-(\pi x^{-1})} * f_{s_2} * f_{h(-x^{-1})} * f_{u_-(\pi x^{-1})} \big),$$

and similarly $f_{s_2} * f_{u_+(x)} * f_{s_2} = c_2^2 f_{u_-(\pi^2 x)}$ *for* $x \in O^\times$.*)*

Furthermore, these relations form a defining set of relations for $\mathcal{H}(G//B_k)$ *as an algebra.*

Remark 6.2. (1) This proposition has an analog for G replaced by $GL(n)$, proved in [HM85], Theorem 2.1, Chapter 3. Unlike the $GL(n)$ case where the length-zero element t lies in $GL(n,F)$ and normalizes B_k, in $SL(2,F)$ the element $t_0 = s_1^{-1} s_2 = \mathrm{diag}(\pi, \pi^{-1})$ has length 2 in W_a and does not normalize B_k. However, the outer automorphism $\mathrm{Ad}(t)$ of $SL(2,F)$ preserves B and B_k and swaps s_1 and s_2, allowing us to use the Coxeter generators $\{s_1, s_2\}$ directly. Note that for $t_0 = h(-1) s_1 s_2$, length additivity gives $f_{t_0} = f_{h(-1)} * f_{s_1} * f_{s_2}$ and $f_{t_0^{-1}} = f_{h(-1)} * f_{s_2} * f_{s_1}$, which satisfy $f_{t_0} * f_{s_1} = f_{s_1} * f_{t_0^{-1}}$.

(2) Note that though S is infinite, the set of generators $\{f_g \mid g \in S\}$ is finite (since f_b for $b \in B$ depends only on the coset $bB_k \in B/B_k$), so the algebra is indeed finitely generated.

(3) Possibly B(iii) is a superfluous relation once restricted to B/B_k.

(4) Note that s_1 and s_2 have order 4 in G ($s_1^2 = s_2^2 = -I \in B$), not order 2 as the analogous permutation matrices in [HM85] do.

(5) The volume constants are computed as follows:

$$\mathrm{vol}(B_k s_1 B_k) = \mathrm{vol}(B_k s_1 B_k s_1^{-1}) / \mathrm{vol}(B_k) = |U_k^- B_k / B_k| = |U_k^- / (U_k^- \cap B_k)| = |\pi^k O / \pi^{k+1} O| = q,$$

and since $t B_k t^{-1} = B_k$ and $t s_1 t^{-1} = s_2$,

$$\mathrm{vol}(B_k s_2 B_k) = \mathrm{vol}(t B_k s_1 B_k t^{-1}) = \mathrm{vol}(B_k s_1 B_k) = q.$$

More generally, for any reduced alternating word $w = s_{i_1} s_{i_2} \cdots s_{i_r}$ ($i_j \in \{1,2\}$, $i_j \neq i_{j+1}$) and any $b \in B$, Iwahori factorization for B_k gives

$$\mathrm{vol}(B_k b w B_k) = \mathrm{vol}(B_k w B_k) = \prod_{j=1}^{r} \mathrm{vol}(B_k s_{i_j} B_k) = q^r,$$

so by Lemma 5.3, $f_w = f_{s_{i_1}} * f_{s_{i_2}} * \cdots * f_{s_{i_r}}$.

Here are some general facts we will use in the proof below.

Lemma 6.3. (a) B_k *is a normal subgroup of* B, *and* $t B_k t^{-1} = B_k$.

(b) *Let* $U_k^+ = \{u_+(x) \mid x \in \pi^k O\}$ *and* $U_k^- = \{u_-(x) \mid x \in \pi^k O\}$. *We have*

$$B_k \cdot (s_1^{-1} B_k s_1) = (s_1^{-1} B_k s_1) \cdot B_k = U_k^- B_k,$$

and

$$B_k \cdot (s_2^{-1} B_k s_2) = (s_2^{-1} B_k s_2) \cdot B_k = U_{k-1}^+ B_k,$$

and both $U_k^- B_k$ *and* $U_{k-1}^+ B_k$ *are groups containing* B_k *as a normal subgroup of index* q.

(c) *In general, we have*

$$(f_{g_1} * f_{g_2})(x) = \int_G f_{g_1}(y) f_{g_2}(y^{-1}x)\, dy = \mathrm{vol}(B_k g_1 B_k \cap x B_k g_2^{-1} B_k).$$

(d) *In general, we have*

$$\int_G (f*g)(x)\, dx = \int_G f(y)\, dy \int_G g(x)\, dx, \quad \int_G (f*g*h)(x)\, dx = \int_G f(x)\, dx \int_G g(y)\, dy \int_G h(z)\, dz.$$

Proof. Parts (a) and (b) may be proven by direct matrix calculations (with the second identity in (b) following from the first by conjugating by t). Parts (c) and (d) are simple consequences of the definitions. □

Proof of Proposition 6.1. We shall follow the ideas in [HM85], proof of Theorem 2.1.

We begin by verifying A(i): $f_{s_1} * f_{s_1} = c_1 \sum_{x \in X} f_x$, where x runs over a complete set X of representatives of $((-1) \cdot (\mathrm{Ad}\, s_1)(B_k) \cdot B_k)/B_k$. As in [HM85], parts (a) and (b) of Lemma 6.3 imply that

$$f_{s_1} * f_{s_1} = c_1 \cdot \chi_{(-1)\cdot U_k^- B_k},$$

where χ_E denotes the characteristic function of a subset E of G and where $c_1 \neq 0$ is some constant. In other words,

$$(f_{s_1} * f_{s_1})(g) = c_1 \sum_{x \in X} f_x(g), \quad g \in G.$$

Plugging $g = -I = s_1^2$ into both sides and using Lemma 6.3(c), we obtain

$$\mathrm{vol}(B_k s_1 B_k) = c_1.$$

Conjugating by the measure-preserving automorphism $\mathrm{Ad}(t)$ (which maps B_k to B_k and s_1 to s_2) immediately gives A(ii) with $c_2 = \mathrm{vol}(B_k s_2 B_k) = q$.

Part (c) of Lemma 6.3 implies part B(i) of the proposition.

We now verify part B(ii). We know $\mathrm{vol}(B_k) = 1$ implies $\mathrm{vol}(bB_k) = 1$, for each $b \in B$. Since B_k is normal in B, $B_k b B_k = bB_k$, so Lemma 5.3 implies part B(ii).

Part B(iii) follows from the definition of f_g.

We shall now verify C(i). Since B_k is normal in B, for any $b \in B$ we have

$$\mathrm{vol}(B_k s_i b B_k) = \mathrm{vol}(B_k s_i b B_k b^{-1} b) = \mathrm{vol}(B_k s_i B_k b) = \mathrm{vol}(B_k s_i B_k).$$

This and Lemma 5.3 imply $f_{s_i} * f_x = f_{s_i x}$, for $x \in B$. Likewise, if $s_i x s_i^{-1} \in B$ then $f_{s_i x s_i^{-1}} * f_{s_i} = f_{s_i x}$. This implies C(i).

We next verify C(ii). The proof of this part is sketched in [HM85]. The argument in [HM85] gives $f_{s_1} * f_{u_+(x)} * f_{s_1} = c f_{u_-(-x)}$, for $x \in O^\times$, for some constant $c \neq 0$. Here c may be evaluated using part (d) of Lemma 6.3, which implies

$$c = \frac{\mathrm{vol}(B_k s_1 B_k)^2 \,\mathrm{vol}(B_k u_+(x) B_k)}{\mathrm{vol}(B_k u_-(-x) B_k)} = \mathrm{vol}(B_k s_1 B_k)^2 = c_1^2.$$

(Here we have used the fact that $u_+(x) \in B$, so $\mathrm{vol}(B_k u_+(x) B_k) = \mathrm{vol}(B_k u_+(x)) = \mathrm{vol}(B_k) = 1$, and the fact that $\mathrm{vol}(B_k u_-(-x) B_k) = \mathrm{vol}(s_1^{-1} B_k s_1 u_+(x) s_1^{-1} B_k s_1) = \mathrm{vol}(s_1^{-1} B_k s_1 u_+(x)) = \mathrm{vol}(s_1^{-1} B_k s_1) = 1$.) Since $u_-(-x) = u_+(-x^{-1}) s_1 h(-x) u_+(-x^{-1})$ with $u_+(-x^{-1}), h(-x) \in B$, we have $f_{u_-(-x)} = f_{u_+(-x^{-1})} * f_{s_1} * f_{h(-x)} * f_{u_+(-x^{-1})}$ by C(i) and B(ii).

Part C(iii) follows directly from C(ii) by applying the automorphism $\mathrm{Ad}(t)$.

It remains to show that this list of relations is a defining set of relations for the algebra. To this end, we follow the proof of [HM85], Chapter 3, Theorem 2.1, as follows.

For $g \in S$, let $\tilde{f}_g$ denote an abstract element. Let $\tilde{\mathcal{H}}$ denote the free algebra generated by these elements modulo the relations (A)–(C) in the proposition. We want to show that the natural homomorphism

$$\tilde{\mathcal{H}} \longrightarrow \mathcal{H}(G//B_k) \tag{6.1}$$

is an isomorphism.

Choose a set of reduced alternating words $w = s_{i_1}s_{i_2}\cdots s_{i_r}$ ($i_j \in \{1,2\}$, $i_j \neq i_{j+1}$) representing the elements of W_a, and define $\tilde{f}_w = \tilde{f}_{s_{i_1}} * \tilde{f}_{s_{i_2}} * \cdots * \tilde{f}_{s_{i_r}}$ (with $\tilde{f}_1$ for the trivial word). Let $\tilde{\mathcal{J}}$ denote the vector space span in $\tilde{\mathcal{H}}$ generated by the elements $\tilde{f}_g = \tilde{f}_x * \tilde{f}_w * \tilde{f}_y$, where $g = xwy$ for $x, y \in B$ and $w \in W_a$. As in [HM85], pages 38–39, the relations (B) and (C) imply that $\tilde{f}_g$ depends only on the double coset $B_k g B_k$, not on the particular representation $g = xwy$. Furthermore, $\tilde{\mathcal{J}}$ is invariant under left and right multiplication by any abstract generator $\tilde{f}_g$, $g \in S$: multiplication by $\tilde{f}_b$ ($b \in B$) is absorbed via B(ii), while multiplication by $\tilde{f}_{s_i}$ either increases the length of w or is reduced via C(i), C(ii), C(iii), and A(i)–(ii) exactly as in [HM85], pages 38–39.

From this invariance property and $\tilde{f}_1 \in \tilde{\mathcal{J}}$, it follows that $\tilde{\mathcal{J}} = \tilde{\mathcal{H}}$. Thus $\tilde{\mathcal{H}}$ is spanned by the elements $\tilde{f}_g$ indexed by the double cosets $B_k g B_k \in B_k\backslash G/B_k$. Since the images $f_g \in \mathcal{H}(G//B_k)$ of distinct double cosets have disjoint supports, they are linearly independent in $\mathcal{H}(G//B_k)$, which implies that the map (6.1) is injective.

Finally, since B_k is normal in B, for any $b_1, b_2 \in B$ and $w \in W_a$ we have

$$\mathrm{vol}(B_k b_1 w b_2 B_k) = \mathrm{vol}(b_1 B_k w B_k b_2) = \mathrm{vol}(B_k w B_k).$$

This and Lemma 5.3 imply $f_{b_1} * f_w * f_{b_2} = f_{b_1 w b_2}$. Combining this with the Iwahori-Bruhat decomposition $G = BW_aB$ and $f_w = f_{s_{i_1}} * \cdots * f_{s_{i_r}}$, we see that $\mathcal{H}(G//B_k)$ is spanned by the image of $\tilde{\mathcal{H}}$, so the map (6.1) is surjective. □

Example 6.4. In the running example with $k = 1$, relation A(i) reads $f_{s_1} * f_{s_1} = 3\sum_x f_x$. By Lemma 6.3(b), the set $(-1)\cdot U_1^- B_1$ is a union of $q = 3$ cosets xB_1, so the sum has three terms. We can check the relation against the trivial representation of Example 3.4, in which f_g acts by $\mathrm{vol}(B_1 g B_1)$. The left side acts by $3 \cdot 3 = 9$. On the right side each x lies in B, so each f_x acts by 1, and the right side acts by $3 \cdot 3 = 9$ as well.

7. The structure of $\mathcal{H}(\overline{G}//B_k)_{gen}$

This section is the metaplectic analog of the previous section. We want to determine the structure of the metaplectic generalized Hecke algebra as a finitely generated algebra with generators and relations.

Throughout this section, we assume that $k > 0$ is chosen so large that $1 + \pi^k O \subset (F^\times)^n$ (for instance, $k > 2\,\mathrm{ord}_\pi(n)$). As noted in Section 4, this guarantees that $(u, y) = 1$ for all $u \in 1 + \pi^k O$ and all $y \in F^\times$, so that $\overline{B}_k$ splits. If the residual characteristic p of F is relatively prime to n, then $1 + \pi O \subset (F^\times)^n$ already holds for $k = 1$, and in fact $\overline{B}$ splits.

Remark 7.1. Before stating Proposition 7.2, let us note how the n-fold cover behaves on the diagonal torus $T = \{h(a) \mid a \in F^\times\}$ and on B:

(1) For $h(a), h(b) \in T$, Kubota's cocycle gives $\beta(h(a), h(b)) = (b, a)$, so the commutator in $\overline{T}$ is $[(h(a), 1), (h(b), 1)] = (b, a)^2 = (b^2, a)$.
(2) When $n = 1$ or $n = 2$ (the classical 2-fold metaplectic cover), $(b, a)^2 = 1$ for all $a, b \in F^\times$, so $\overline{T}$ is abelian. Moreover, when $1 + \pi^k O \subset (F^\times)^n$, for any $n \geq 1$ the subgroup $T_k = T \cap B_k = \{h(u) \mid u \in 1 + \pi^k O\}$ satisfies $(u, a) = 1$ for all $a \in F^\times$, so $\overline{T}_k$ lies in the center of $\overline{T}$.

(3) For general $n \geq 1$, let $B^{(n)} = \{b \in B \mid$ the diagonal entries of b lie in $(O^\times)^n(1+\pi^k O)\}$. On $B^{(n)}$, all diagonal entries are n^{th} powers in $F^\times$, so the Hilbert symbol vanishes identically on $B^{(n)}$ and $\overline{B^{(n)}}$ splits and commutes with $(t_0, 1)$. In addition, choosing coset representatives of $B/B^{(n)} \cong O^\times/(O^\times)^n$ and of the double cosets $B_k \backslash G / B_k$, we may choose the coboundary $s : G \to \mu_n$ in Lemma 5.7 (with $s|_{B_k} \equiv 1$) so that β' trivializes the cocycle on the chosen coset decompositions.

Let the Haar measure on $\overline{G}$ be normalized so that $\overline{B}_k$ has measure one. Convolution gives $\mathcal{H}(\overline{G}//B_k)_{gen}$ the structure of an algebra.

Proposition 7.2. *Assume that* $\mathrm{vol}(\overline{B}_k) = 1$, *that* $k > 0$ *is chosen so large that* $1 + \pi^k O \subset (F^\times)^n$, *and choose the equivalent cocycle* β' *via Lemma 5.7 as above. The algebra* $\mathcal{H}(\overline{G}//B_k)_{gen}$ *is generated by the functions* $\overline{f}_g$, $g \in S$. *On the* n^{th}*-power Iwahori subgroup* $B^{(n)}$ *(and on all of* B *when* $n \mid 2$ *and* $\gcd(p, n) = 1$, *or under the normalized coset splitting of Remark 7.1), these elements are subject to the same relations as in Proposition 6.1, except that* f_g *must be replaced by* $\overline{f}_g$. *Furthermore, these relations form a defining set of relations for* $\mathcal{H}(\overline{G}//B_k)_{gen}$ *as an algebra.*

Proof. First we verify that the claimed relations hold.

A(i), A(ii): By Lemma 5.6, we have

$$(\overline{f}_{s_i} * \overline{f}_{s_i})(x, 1) = \int_{B_k s_i B_k} \overline{f}_{s_i}(y^{-1}x, 1)\beta'(y, y^{-1}x)^{-1}\, dy.$$

Because $1 + \pi^k O \subset (F^\times)^n$, β' is trivial on $U_k^- B_k$ and $U_{k-1}^+ B_k$, and for the same reason as in the non-metaplectic case, $\overline{f}_{s_i} * \overline{f}_{s_i}$ is supported on $\overline{((-1) \cdot (\mathrm{Ad}\, s_i)(B_k) \cdot B_k)}$. Evaluating at $(-I, 1)$ gives the same constant $c_i = \mathrm{vol}(B_k s_i B_k) = q$ as in the non-metaplectic case, which proves A(i) and A(ii).

B(i): For $g \in \overline{G}$ and $\phi \in \mathcal{H}(\overline{G}//B_k)_{gen}$, since $\phi(gx^{-1})\overline{f}_1(x) = \phi(g)$ for all $x = (b, \zeta) \in \overline{B}_k$ (by right B_k^*-invariance and genuineness of ϕ), we have

$$(\phi * \overline{f}_1)(g) = \int_{\overline{G}} \phi(gx^{-1})\overline{f}_1(x)\, dx = \int_{\overline{B}_k} \phi(gx^{-1})\overline{f}_1(x)\, dx = \phi(g)\,\mathrm{vol}(\overline{B}_k) = \phi(g).$$

Similarly, $\overline{f}_1 * \phi = \phi$, so $\overline{f}_1$ is the identity in $\mathcal{H}(\overline{G}//B_k)_{gen}$.

B(ii): By Lemma 5.6, we have

$$(\overline{f}_x * \overline{f}_y)(g, 1) = \int_{B_k x B_k} \overline{f}_y(h^{-1}g, 1)\beta'(h, h^{-1}g)^{-1}\, dh.$$

This is non-zero if and only if $gy^{-1} = xb$ for some $b \in B_k$, since B_k is normal in B. This implies $\mathrm{supp}(\overline{f}_x * \overline{f}_y) \subset \overline{B_k x y B_k}$, and Lemma 5.5 yields $\overline{f}_x * \overline{f}_y = \overline{f}_{xy}$.

B(iii): This is an immediate consequence of Lemma 5.5.

C(i): The proof of this identity is analogous to the proof in the non-metaplectic case (using Lemma 5.5 in place of Lemma 5.3 where appropriate) and is omitted.

C(ii), C(iii): These are proven in the same way as in the non-metaplectic case, so we only sketch the argument. The proof in the non-metaplectic case shows that each of these triple convolutions is supported on a single double $\overline{B}_k$-coset. On the other hand, the support of each non-zero function in the Hecke algebra contains at least one double $\overline{B}_k$-coset, since different double $\overline{B}_k$-cosets are disjoint. Since convolution preserves genuineness and bi-B_k^*-invariance, those functions whose support is precisely one double $\overline{B}_k$-coset are constant multiples of the corresponding $\overline{f}_g$. The constant multiple

is evaluated at the double coset representative exactly as in the non-metaplectic case, and since the measures differ only by the normalized measure on μ_n, the volume computations are identical.

It remains to show that this list of relations is a defining set of relations for the algebra. Since $\{\overline{f}_g \mid B_k g B_k \in B_k\backslash G/B_k\}$ forms a vector space basis of $\mathcal{H}(\overline{G}//B_k)_{gen}$ indexed by the same double cosets as $\mathcal{H}(G//B_k)$, the argument is identical to the non-metaplectic case in Proposition 6.1. □

Example 7.3. In the running example, $n = 2$ divides 2 and $\gcd(p,n) = \gcd(3,2) = 1$. So Proposition 7.2 applies with $k = 1$, and its relations hold on all of B, not only on $B^{(2)}$. For instance, relation A(i) becomes $\overline{f}_{s_1} * \overline{f}_{s_1} = 3\sum_x \overline{f}_x$, with the same three representatives x as in Example 6.4.

8. Application to representation theory

This section combines Propositions 6.1 and 7.2 into the main theorem, an isomorphism of the two Hecke algebras. It then translates the isomorphism into a correspondence of representations (Corollary 8.2). Finally it shows that the correspondence preserves supercuspidal and square-integrable representations (Corollary 8.12). The background for this section is in Section 3.4.

The following is our main result:

Theorem 8.1. *If $k > 0$ is chosen sufficiently large as in Proposition 7.2 (so in particular $\overline{B}_k$ splits), then the map $f_g \longmapsto \overline{f}_g$, $g \in S$, defines an isomorphism of algebras $\mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$.*

Proof. This is an immediate consequence of Propositions 6.1 and 7.2. □

Corollary 8.2. *If $k > 0$ is as in Theorem 8.1, then the algebra isomorphism $\eta : \mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$ induces a set-theoretic bijection $\eta^* : \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} \to \mathcal{H}(G//B_k)^\wedge$ on the sets of equivalence classes of irreducible (genuine, in the case of $\overline{G}$) representations. Furthermore, if*

$$\begin{array}{ccc} \mathcal{H}(G//B_k) & \longrightarrow & \mathcal{H}(\overline{G}//B_k)_{gen} \\ f & \longmapsto & \overline{f} \end{array}$$

and

$$\begin{array}{ccc} \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} & \longrightarrow & \mathcal{H}(G//B_k)^\wedge \\ \overline{\pi} & \longmapsto & \pi \end{array}$$

then $\mathrm{tr}(\overline{\pi}(\overline{f})) = \mathrm{tr}(\pi(f))$.

Proof. Let $\eta : \mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$ be the algebra isomorphism of Theorem 8.1, and let $\eta^* : \mathcal{H}(\overline{G}//B_k)^\wedge_{gen} \to \mathcal{H}(G//B_k)^\wedge$ be the pullback map defined by $\eta^*(\overline{\pi}) = \overline{\pi} \circ \eta$. Suppose $\pi = \eta^*(\overline{\pi})$ and $\pi' = \eta^*(\overline{\pi}')$ are equivalent in $\mathcal{H}(G//B_k)^\wedge$. Since η is surjective, every $\overline{f} \in \mathcal{H}(\overline{G}//B_k)_{gen}$ is of the form $\overline{f} = \eta(f)$ for some $f \in \mathcal{H}(G//B_k)$, so any intertwining isomorphism between π and π' also intertwines $\overline{\pi}$ and $\overline{\pi}'$. Thus η^* is injective. Conversely, for any $(\pi, W) \in \mathcal{H}(G//B_k)^\wedge$, the composition $\overline{\pi} = \pi \circ \eta^{-1}$ defines an irreducible representation in $\mathcal{H}(\overline{G}//B_k)^\wedge_{gen}$ satisfying $\eta^*(\overline{\pi}) = \pi$, so η^* is surjective.

The equality of the traces $\mathrm{tr}(\overline{\pi}(\eta(f))) = \mathrm{tr}(\pi(f))$ holds by definition of η^*. □

Example 8.3. In the running example with $k = 1$, Theorem 8.1 gives an isomorphism $\mathcal{H}(SL(2,\mathbb{Q}_3)//B_1) \to \mathcal{H}(\overline{G}//B_1)_{gen}$ with $f_g \mapsto \overline{f}_g$. Example 3.4 gives the one-dimensional module of $\mathcal{H}(G//B_1)$ in which f_g acts by $\mathrm{vol}(B_1 g B_1)$. Under the bijection of Corollary 8.2, it corresponds to the one-dimensional module of $\mathcal{H}(\overline{G}//B_1)_{gen}$ in which each generator $\overline{f}_g$, $g \in S$, acts by the same number $\mathrm{vol}(B_1 g B_1)$. For example, both f_{s_1} and $\overline{f}_{s_1}$ act by 3, so the traces in Corollary 8.2 agree.

Definition 8.4. Let K denote a compact open subgroup of $\overline{G}$. A locally constant complex-valued function f on $\overline{G}$ is called **left** (resp., **right**) **Hecke finite** if the vector space spanned by all functions of the form $h * f$ (resp., $f * h$), for $h \in \mathcal{H}(\overline{G}//K)$, is finite dimensional. Let $\mathcal{A}_{gen}(\overline{G})$ denote the space of all locally constant complex-valued genuine functions f on $\overline{G}$ for which

- f is double K-finite,
- f is right- and left-Hecke finite.

This space is called the space of **matrix coefficients** of $\overline{G}$. If (π, V) is a genuine admissible representation of $\overline{G}$ then the span of the elements $f(g) = \langle v_0^*, \pi(g)v_0 \rangle$ $(g \in \overline{G})$, for all $v_0 \in V$ and $v_0^* \in V^*$, the smooth contragredient of V, is called the space of **matrix coefficients of** π and is denoted $\mathcal{A}_{gen}(\pi)$.

Example 8.5. Take $n = 1$ and the trivial representation $\mathbf{1}$ of G. Its matrix coefficients $g \mapsto \langle v_0^*, v_0 \rangle$ are the constant functions. So $\mathcal{A}(\mathbf{1})$ is the one-dimensional space of constants. When $n \geq 2$ a nonzero constant is not genuine (Example 4.4), which reflects the fact that the matrix coefficients of a genuine representation are genuine functions.

Let ρ (resp., λ) denote the right (resp., left) regular representation of $\overline{G}$ on $\mathcal{A}_{gen}(\overline{G})$. The next proposition establishes that the metaplectic analogs of (the relevant parts of) Corollary 1.10.6 in [Si79] for $\overline{G}$ are true. The result below implies that we can "explicitly" realize the space of any genuine irreducible representation of $\overline{G}$ as a subspace of $\mathcal{A}_{gen}(\overline{G})$, a fact we will make use of later.

Proposition 8.6. *Let $f \in \mathcal{A}_{gen}(\overline{G})$ and let V_f denote the $\overline{G}$-module spanned by the translates $\rho(g)f$, $g \in \overline{G}$, and let π denote the restriction of ρ to V_f. We have:*

(a) *There exist elements $v_0 \in V_f$ and $v_0^* \in V_f^*$, the contragredient of V_f, such that $f(g) = \langle v_0^*, \pi(g)v_0 \rangle$, for all $g \in \overline{G}$.*

(b) *$\mathcal{A}_{gen}(\pi)$ is spanned by $\lambda(x)\rho(y)f$, $x, y \in \overline{G}$.*

Proof. The proof of this follows from the general arguments given in [Si79]. The results in that part of the book [Si79] hold for more general totally disconnected groups, so one need only check that the argument remains valid when "genuineness" is also assumed. □

We need to recall some facts regarding the relationship between smooth representations of $\overline{G}$ and finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-modules. To this end, we recall briefly how one can construct smooth representations of $\overline{G}$ from finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-modules—both the so-called "induced" and "produced" modules in §2 of Borel [Bo76] will do this job. The "inverse" construction is then given along with some of the basic properties of these constructions in Proposition 8.9 below.

Definition 8.7. If (r, W) is a finite dimensional $\mathcal{H}(\overline{G}//B_k)$-module then we define $(I(r), I(W))$ to be the **induced** $\overline{G}$-module and $(P(r), P(W))$ to be the (smooth) **produced** $\overline{G}$-module constructed in [Bo76]. Analogously, if (r, W) is a finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module then we define $(I_{gen}(r), I_{gen}(W))$ to be the **induced** genuine $\overline{G}$-module and $(P_{gen}(r), P_{gen}(W))$ to be the **produced** genuine $\overline{G}$-module obtained by replacing $C_c(\overline{G}/B_k)$ by $C_c(\overline{G}/B_k)_{gen}$ in the constructions in [Bo76].

Example 8.8. Take $n = 1$, and let $W = \mathbb{C}$ be the module of Example 3.4, on which $f \in \mathcal{H}(G//B_k)$ acts by $\int_G f$. It is irreducible, being one-dimensional. By Proposition 8.9(d) (in the case $n = 1$), the induced module $I(W)$ is irreducible. Take $V = \mathbf{1}$ in Proposition 8.9(c). Then $V^{B_k} = \mathbb{C}$ is the module W itself, so $\mathrm{Hom}_G(I(W), \mathbf{1}) \cong \mathrm{Hom}_{\mathcal{H}(G//B_k)}(W, W) \neq 0$. A nonzero map between irreducible representations is an isomorphism. So $I(W)$ is isomorphic to the trivial representation $\mathbf{1}$.

Proposition 8.9. *Let (r, W) be a finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module and let (π, V) be a genuine smooth representation of $\overline{G}$.*

(a) *There are natural $\mathcal{H}(\overline{G}//B_k)_{gen}$-module isomorphisms*

$$W \cong I_{gen}(W)^{B_k} \quad \text{and} \quad W \cong P_{gen}(W)^{B_k}.$$

(b) *$I_{gen}(W)$ is generated, as a $\overline{G}$-module, by $I_{gen}(W)^{B_k}$.*

(c) *The natural restriction maps yield isomorphisms*

$$\rho_P : \mathrm{Hom}_{\overline{G}}(V, P_{gen}(W)) \longrightarrow \mathrm{Hom}_{\mathcal{H}(\overline{G}//B_k)_{gen}}(V^{B_k}, W)$$

and

$$\rho_I : \mathrm{Hom}_{\overline{G}}(I_{gen}(W), V) \longrightarrow \mathrm{Hom}_{\mathcal{H}(\overline{G}//B_k)_{gen}}(W, V^{B_k}).$$

(d) *(r, W) is an irreducible $\mathcal{H}(\overline{G}//B_k)_{gen}$-module if and only if $(I_{gen}(r), I_{gen}(W))$ is an irreducible representation of $\overline{G}$. Moreover, if (r, W) is an irreducible $\mathcal{H}(\overline{G}//B_k)_{gen}$-module then $I_{gen}(r) \cong P_{gen}(r)$.*

Proof. In the case $n = 1$, the statements in the proposition above are either in §§1–2 of [Bo76] as stated or follow from Borel's results using Schur's lemma (see, for example, Lemma 3.14 in [CR81]). In general, one need only check that the argument remains valid when "genuineness" is also assumed. □

Definition 8.10. We define a genuine admissible irreducible representation of $\overline{G}$ to be **supercuspidal** if and only if its matrix coefficients are compactly supported. We define a genuine admissible irreducible representation of $\overline{G}$ to be **square-integrable** if and only if its matrix coefficients belong to $L^2(\overline{G})$.

We define an irreducible finite dimensional $\mathcal{H}(\overline{G}//B_k)_{gen}$-module (r, W) to be **supercuspidal** (resp., **square-integrable**) if and only if $I_{gen}(r)$ is a supercuspidal (resp., square-integrable) representation of $\overline{G}$.

Example 8.11. For $n = 1$, the trivial representation of $G = SL(2, F)$ is neither supercuspidal nor square-integrable (Example 3.4). The Steinberg representation of $SL(2, F)$ is square-integrable but not supercuspidal. It has a nonzero Iwahori-fixed vector, and a representation with a nonzero Iwahori-fixed vector is a subquotient of an unramified principal series, so it is not supercuspidal. By Example 8.8, the trivial $\mathcal{H}(G//B_k)$-module is neither supercuspidal nor square-integrable in the sense of the second paragraph of Definition 8.10.

Next we observe that, as in the case of $GL(n)$, the correspondence of Corollary 8.2 preserves supercuspidals and discrete series representations ([HM85], page 33).

Corollary 8.12. *Assume $k > 0$ is as in Corollary 8.2.*

- *An admissible genuine representation $\overline{\pi} \in \mathcal{H}(\overline{G}//B_k)^{\wedge}_{gen}$ is supercuspidal if and only if π is, where $\pi, \overline{\pi}$ are as in Corollary 8.2.*
- *$\overline{\pi}$ is square-integrable if and only if π is, where $\pi, \overline{\pi}$ are as in Corollary 8.2.*

Proof. Suppose that the mapping $\mathcal{H}(\overline{G}//B_k)^{\wedge}_{gen} \to \mathcal{H}(G//B_k)^{\wedge}$ sends a representation $(\overline{r}, \overline{W})$ to a representation (r, W). Note that the construction of the correspondence in Corollary 8.2 implies that $W = \overline{W}$. From the construction of $I_{gen}(\overline{W}) = C_c(\overline{G}/B_k)_{gen} \otimes_{\mathcal{H}(\overline{G}//B_k)_{gen}} \overline{W}$ (more precisely, from the definition of $\otimes_{\mathcal{H}(\overline{G}//B_k)_{gen}}$), we see that the $\overline{r}(\overline{f}_g)$-action on $\overline{W} \cong I_{gen}(\overline{W})^{B_k}$ corresponds

to the $\overline{f}_g$-action (via convolution) on $C_c(\overline{G}/B_k)_{gen}$. Identifying $\overline{W}$ with $I_{gen}(\overline{W})^{B_k}$, we see that each element of $\overline{W}$ is Hecke-finite and may therefore be regarded as a matrix coefficient of $\overline{G}$. Let $(\overline{\pi}, \overline{V})$ denote the $\overline{G}$-module generated by an element $w \in \overline{W} \cong I_{gen}(\overline{W})^{B_k}$. By Proposition 8.6 (or Proposition 8.9(b)), $\overline{\pi} = I_{gen}(\overline{r})$.

From this and the fact that $W = \overline{W}$, it follows that if the elements of $\overline{W}$ are compactly supported (when regarded as matrix coefficients of $\overline{G}$) then the elements of W are also compactly supported (when regarded as matrix coefficients of G). Now assume that $(\overline{r}, \overline{W})$ is supercuspidal, so the matrix coefficients of $I_{gen}(\overline{r})$ have compact support. As indicated above, (r, W) is supercuspidal as well, and the converse holds symmetrically. This establishes the first part of the corollary.

The second part of the corollary is proved similarly, hence is omitted. □

Example 8.13. In the running example, the genuine module of Example 8.3 corresponds to the trivial representation of $SL(2, \mathbb{Q}_3)$. By Example 8.11 the trivial representation is neither supercuspidal nor square-integrable. So Corollary 8.12 shows that the genuine representation $I_{gen}(W)$ of $\overline{G}$ attached to this module is neither supercuspidal nor square-integrable.

9. The Iwahori algebra

This section treats the level $k = 0$, where B_k is replaced by the Iwahori subgroup B itself. The method of the earlier sections needs $k > 0$. At level 0 the genuine Hecke algebra is instead described by a theorem of Savin, which compares $\overline{G}$ with a linear group that depends on the parity of n.

In the case where $k = 0$ and $\gcd(p, 2n) = 1$, the genuine Iwahori-Hecke algebra $\mathcal{H}(\overline{G}//B)_{gen}$ is described by a general theorem due to Savin [Sa88]. Recall that for $G = SL(2, F)$, the dual group of the n-fold metaplectic cover $\overline{G}$ depends on the parity of n: let $G^\vee = SL(2, F)$ when n is odd, and let $G^\vee = PGL(2, F) \cong SO(3, F)$ when n is even (with Iwahori subgroup $B^\vee \subset G^\vee$). We recall Savin's theorem in this setting:

Theorem 9.1 (Savin [Sa88])**.** *Assume* $\gcd(p, 2n) = 1$. *There is an isomorphism of Iwahori-Hecke algebras*

$$\mathcal{H}(\overline{G}//B)_{gen} \xrightarrow{\sim} \mathcal{H}(G^\vee//B^\vee),$$

where $G^\vee = G = SL(2, F)$ *when* n *is odd, and* $G^\vee = PGL(2, F)$ *when* n *is even.*

Corollary 9.2. *Assume* $\gcd(p, 2n) = 1$. *The isomorphism* $\mathcal{H}(\overline{G}//B)_{gen} \to \mathcal{H}(G^\vee//B^\vee)$ *induces a set-theoretic bijection* $\mathcal{H}(\overline{G}//B)^\wedge_{gen} \to \mathcal{H}(G^\vee//B^\vee)^\wedge$ *on the sets of equivalence classes of irreducible (genuine, in the case of* $\overline{G}$*) representations with a* B*-fixed (resp.,* $B^\vee$*-fixed) vector. Furthermore, if*

$$\begin{array}{ccc} \mathcal{H}(G^\vee//B^\vee) & \longrightarrow & \mathcal{H}(\overline{G}//B)_{gen} \\ f & \longmapsto & \overline{f} \end{array}$$

and

$$\begin{array}{ccc} \mathcal{H}(\overline{G}//B)^\wedge_{gen} & \longrightarrow & \mathcal{H}(G^\vee//B^\vee)^\wedge \\ \overline{\pi} & \longmapsto & \pi \end{array}$$

then $\mathrm{tr}(\overline{\pi}(\overline{f})) = \mathrm{tr}(\pi(f))$.

Example 9.3. In the running example, $\gcd(p, 2n) = \gcd(3, 4) = 1$ and $n = 2$ is even. So Theorem 9.1 identifies $\mathcal{H}(\overline{G}//B)_{gen}$ with the Iwahori–Hecke algebra of $G^\vee = PGL(2, \mathbb{Q}_3)$. In that algebra, with the Iwahori subgroup of volume 1, the characteristic function T_s of the double coset of a simple

reflection s satisfies the quadratic relation $T_s * T_s = (q-1)T_s + qT_1$ of Iwahori and Matsumoto. With $q = 3$ this reads $T_s * T_s = 2T_s + 3T_1$.

10. Summary of results

This section collects the results of the paper in one place and lists the questions it leaves open. Let F be a p-adic field of characteristic 0 with $|\mu_n(F)| = n$, let $G = SL(2,F)$, and let $\overline{G}$ be its n-fold Kubota cover.

(1) Proposition 6.1 gives generators and defining relations for the generalized Hecke algebra $\mathcal{H}(G//B_k)$, $k > 0$.
(2) Proposition 7.2 gives the same generators and relations for the genuine algebra $\mathcal{H}(\overline{G}//B_k)_{gen}$, under the hypotheses stated there, when $1 + \pi^k \mathcal{O} \subset (F^\times)^n$.
(3) Theorem 8.1 deduces that $f_g \mapsto \overline{f}_g$, $g \in S$, is an algebra isomorphism $\mathcal{H}(G//B_k) \to \mathcal{H}(\overline{G}//B_k)_{gen}$.
(4) Corollary 8.2 turns this into a bijection between irreducible modules of the two algebras that preserves traces. Corollary 8.12 shows that the bijection preserves supercuspidal and square-integrable representations.
(5) At the Iwahori level, Theorem 9.1 (due to Savin [Sa88]) and Corollary 9.2 give the analogous statements, with G replaced by $G^\vee$, when $\gcd(p, 2n) = 1$.

Two questions remain open. The first is raised in Remark 6.2: whether relation B(iii) is superfluous once it is restricted to B/B_k. The second is the hope stated in Section 2.1: that these results, with Waldspurger's multiplicity one theorem [Wa80] and the Arthur–Selberg trace formula, may contribute to a proof of Labesse's multiplicity one conjecture for $SL(2)$.

11. Glossary

This section gives short definitions of the technical terms used in the paper. Section 3 explains most of them in more detail.

Admissible representation: A smooth representation whose space of vectors fixed by any compact open subgroup is finite dimensional.

Affine Weyl group: The group $W_a = N_G(T)/(T \cap B)$. For $SL(2,F)$ it is infinite dihedral, generated by the images of s_1 and s_2.

Cocycle (2-cocycle): A function $\alpha : H \times H \to \mu_n$ that makes $H \times \mu_n$ a group under $(h_1, \zeta_1)(h_2, \zeta_2) = (h_1 h_2, \alpha(h_1, h_2)\zeta_1\zeta_2)$.

Filtration subgroup: The normal subgroup B_k of the Iwahori subgroup defined in Section 4.

Generalized Hecke algebra: The convolution algebra of locally constant, compactly supported, bi-K-invariant functions (Definition 4.1).

Genuine function or representation: One on which μ_n acts through the fixed embedding $\mu_n \subset \mathbb{C}^\times$.

Hilbert symbol: The nth power norm residue symbol $F^\times \times F^\times \to \mu_n$.

Induced and produced modules: Borel's two constructions of a representation from a Hecke algebra module (Definition 8.7).

Iwahori subgroup: The subgroup B of $SL(2,\mathcal{O})$ of matrices whose lower left entry lies in $\pi\mathcal{O}$.

Iwahori–Bruhat decomposition: The decomposition $G = BW_aB$ of G into double cosets of B.

Kubota cocycle: The explicit cocycle β of Section 1 defining the n-fold cover of $SL(2,F)$.

Matrix coefficient: A function $g \mapsto \langle v^*, \pi(g)v \rangle$ (Definition 8.4).

Metaplectic cover: The topological central extension $\overline{G}$ of G by μ_n defined by the Kubota cocycle.

Splitting: A function s with $\alpha(h_1, h_2) = s(h_1)s(h_2)s(h_1h_2)^{-1}$. A cover splits when such an s exists.

Square-integrable representation: One whose matrix coefficients are square-integrable (Definition 8.10).

Supercuspidal representation: One whose matrix coefficients are compactly supported (Definition 8.10).

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